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Impact Statement

This paper presents work whose goal is to provide theoretical analysis and practical tools to address the data contamination issue caused by machine-generated content. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Mathematical Theory: The Proof of Theorem 4.3

In this appendix, we provide a full account of the mathematical details of the theorems and their proofs appearing in the main body of the paper. Our proof technique has the same framework as (Bertrand et al., 2024) because our theoretical analysis generalizes theirs to the case where you have a self-correction function in the self-consuming loop.

A.1. Mathematical Setup and Notation

Definition A.1. Define the optimal model parameters to be

$$\theta^\star \in \arg\max_{\theta' \in \Theta} \mathbb{E}_{x \sim p_{\text{data}}}[\log p_{\theta'}(x)], \tag{9}$$

chosen so that $\|\theta^\star\|$ has minimal norm within this set. Let θ be any model parameters. Then the *correction of strength* γ of distribution p_θ towards $p_{\theta^\star}$ is a new distribution, denoted $\pi_\gamma p_\theta$, defined according to the rule

$$\pi_\gamma p_\theta(x) := \frac{p_\theta(x) + \gamma p_{\theta^\star}(x)}{1+\gamma}.$$

This is illustrated in Figure 6. Let θ_t be the parameters of the model trained after t generations. We define the *iterative fine-tuning with correction update mapping* to be

$$\pi_\gamma \mathcal{G}_\lambda^\infty(\theta) := \text{local}\operatorname*{argmax}_{\theta' \in \Theta} \mathcal{H}(\theta, \theta') := \text{local}\operatorname*{argmax}_{\theta' \in \Theta} [\mathbb{E}_{x \sim p_{\text{data}}}[\log p_{\theta'}(x)]] + \lambda \mathbb{E}_{x \sim \pi_\gamma p_\theta}[\log p_{\theta'}(x)]] \tag{10}$$

$$\pi_\gamma \mathcal{G}_\lambda^n(\theta) := \text{local}\operatorname*{argmax}_{\theta' \in \Theta} \hat{\mathcal{H}}(\theta, \theta') := \text{local}\operatorname*{argmax}_{\theta' \in \Theta} [\mathbb{E}_{x \sim \hat{p}_{\text{data}}}[\log p_{\theta'}(x)]] + \lambda \mathbb{E}_{x \sim \widehat{\pi_\gamma p_\theta}}[\log p_{\theta'}(x)]]. \tag{11}$$

Notice that in the finite case, we're optimizing by taking samples from an empirical distribution. In contrast, in the infinite case, there is zero statistical error, since the parameter update is done with access to an infinite sampling budget at each generation t. The finite case is the more practical case, when we have some statistical error (so we only have access to finite sampling at each generation). Since the parameter space of the generative model class might be limited, there might be a small difference between the distribution corresponding to the optimal parameters and the target distribution p_{data}; we capture this difference via the Wasserstein-2 distance and denote

$$\varepsilon := d_W(p_{\theta^\star}, p_{\text{data}}). \tag{12}$$

Let

$$\mathcal{H}_1(\theta') := \mathbb{E}_{x \sim p_{\text{data}}}[\log p_{\theta'}(x)], \qquad \mathcal{H}_2(\theta, \theta') := \mathbb{E}_{x \sim \pi_\gamma p_\theta}[\log p_{\theta'}(x)]. \tag{13}$$

and note that $\mathcal{H}(\theta, \theta') = \mathcal{H}_1(\theta') + \lambda \mathcal{H}_2(\theta, \theta')$.

We first establish that the correction map is truly a mapping of probability distributions as well as some of its elementary properties.

Lemma A.2. *The correction map has the following properties.*

1. *$\pi_\gamma p_\theta$ is a probability distribution.*
2. *Strengths $0, 1, \infty$ correspond to p_θ, the average of p_θ and $p_{\theta^\star}$, and $p_{\theta^\star}$, respectively.*
3. *For any $x \in \mathbb{R}^n$, if $\gamma > 1$, then*
$$\|\pi_\gamma p_\theta(x) - p_{\theta^\star}(x)\| \leq \|\pi_\gamma p_\theta(x) - p_\theta(x)\|,$$
and if $\gamma < 1$, then the inequality is flipped. In other words, $\pi_\gamma p_\theta$ is a better estimate of the ideal distribution $p_{\theta^\star}$ than p_θ is, precisely when the projection strength is more than 1.

Proof. For the first point, $\pi_\gamma p_\theta$ is a probability distribution because it is a convex combination of probability distributions. For example, we can compute that

$$\int_{\mathbb{R}^d} \pi_\gamma p_\theta dx = \frac{1}{1+\gamma} \int_{\mathbb{R}^d} p_\theta(x) dx + \frac{\gamma}{1+\gamma} \int_{\mathbb{R}^d} p_{\theta^\star}(x) dx = \frac{1}{1+\gamma} \cdot 1 + \frac{\gamma}{1+\gamma} \cdot 1 = 1.$$